

What I learned about networking so far

TCP/TCP Handshake:

TCP is a widely used protocol for communication between devices. It is well known for its safety. This safety can be attributed to the fact that it checks after any action done (connecting, sending packets, etc.).

One of the ways that it ensures safety is by the **3 way handshake**.

3 way handshake works like this:

- The device '**A**' sends an ISN which is a randomly generated number. (SYN)
- The device '**B**' receives the ISN, to show that it has received the ISN it adds 1 to it and sends it back along with its own ISN which is also random. (SYN-ACK)
- **A** receives the **B**'s ISN and its own ISN + 1 and sends back **B**'s ISN + 1. (ACK)
- Finally **B** receives the packets and they start transferring files or whatever they do. (ACK)

The TCP header is 20-60 bytes. The 40 bytes are for options and padding so a header that is 20 bytes has no options or padding.

- **The source port address** is 16 bits and holds the port address of application sending data segment.
- **The destination port address** is 16 bits and holds the port address of the application receiving the data segment.
- **Sequence Number** is 32 bits and holds the sequence number used at reassembling the message received which are out of order.
- **Acknowledgement number** is 32 bits and holds acknowledgement. It is ACK of previous bytes received successfully (ISN + 1).
- **Header length** is 4 bits and indicates the length of the header.
- **Control flags** are 6 in number and 1 bit. They control connection establishment, connection termination and connection abortion, mode of transfer and flow control etc.
- **URG** – urgent pointer is valid
- **ACK** – acknowledgement number is valid
- **PSH** – Request to push
- **RST** – Connection reset
- **SYN** – Synchronize sequence number
- **FIN** – Connection termination
- **Window size** – is size of window sending TCP bytes

- **Checksum** – is used for error control and mandatory here
- **Urgent** pointer – is used to point to data that is urgently required which needs to reach the receiving process urgently.

What is UDP

From what I understand it is TCP without the checks like the 3 way handshake.

What is a socket

A socket is an endpoint for two devices to communicate through.

What is a port

A port is a number in which the OS uses to identify the process or service. This way other devices can make requests/send packets through those ports. A port is an unsigned 16 bit integer. When sending a packet the destination address is the IP of the destination device combined with the port number the package will be sent to.

It is important to understand that a socket is not a port. A socket is just for two way communicating. There can be multiple sockets in a port.

So the gist is that when a socket opens at say port 53 (which is associated with TCP in Linux) the process that is connected to the socket is saying to the OS "Any TCP packets arrive to this port direct them to me".